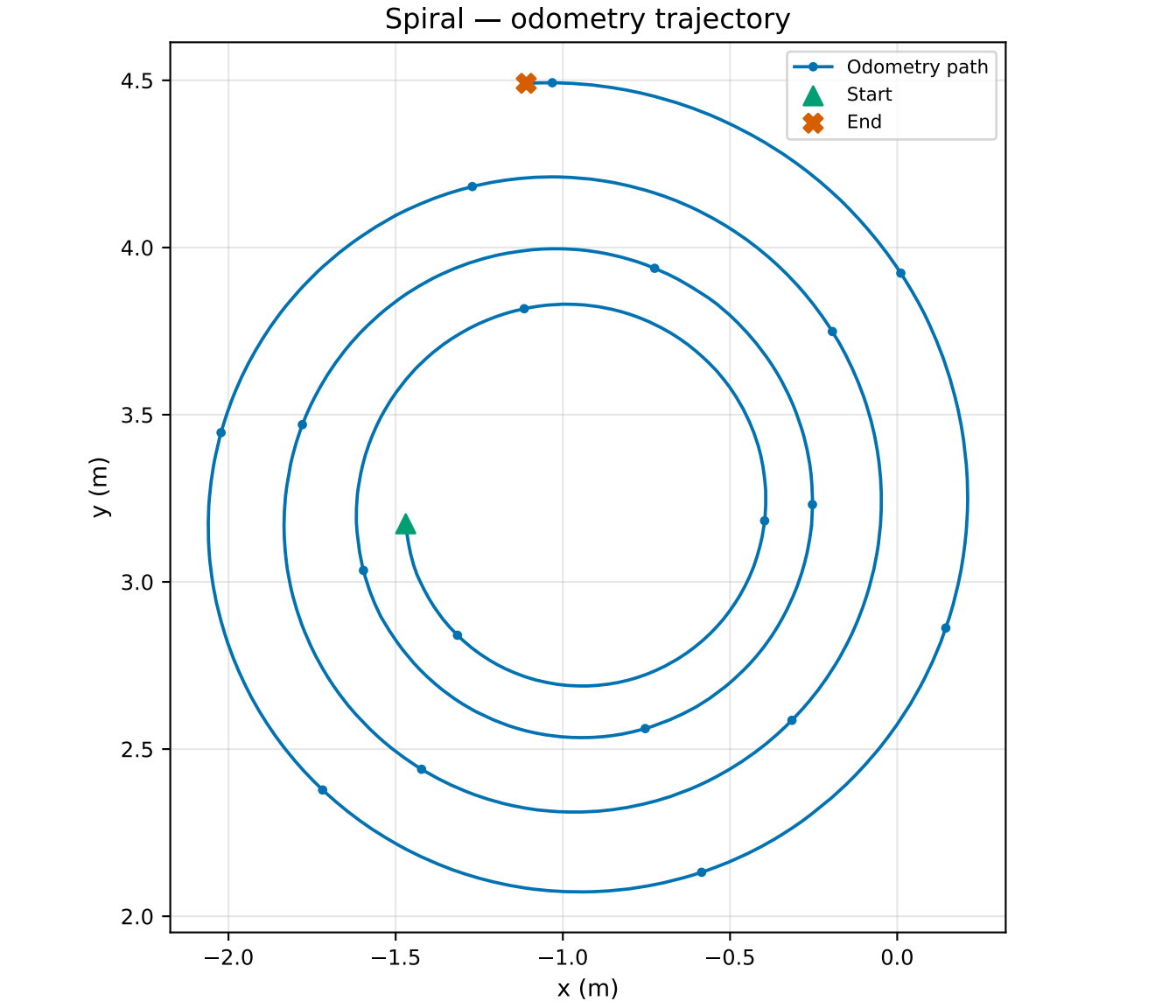
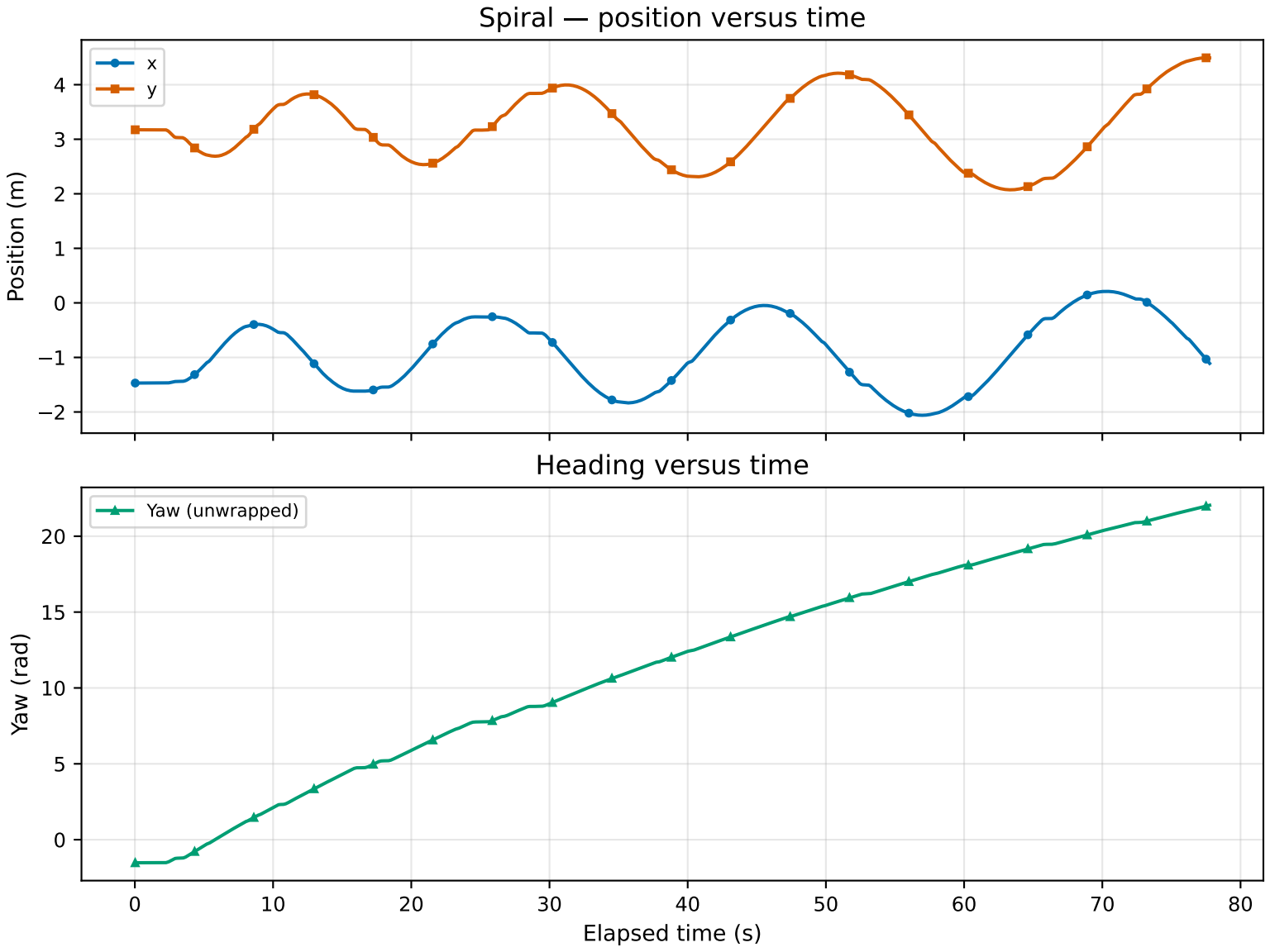
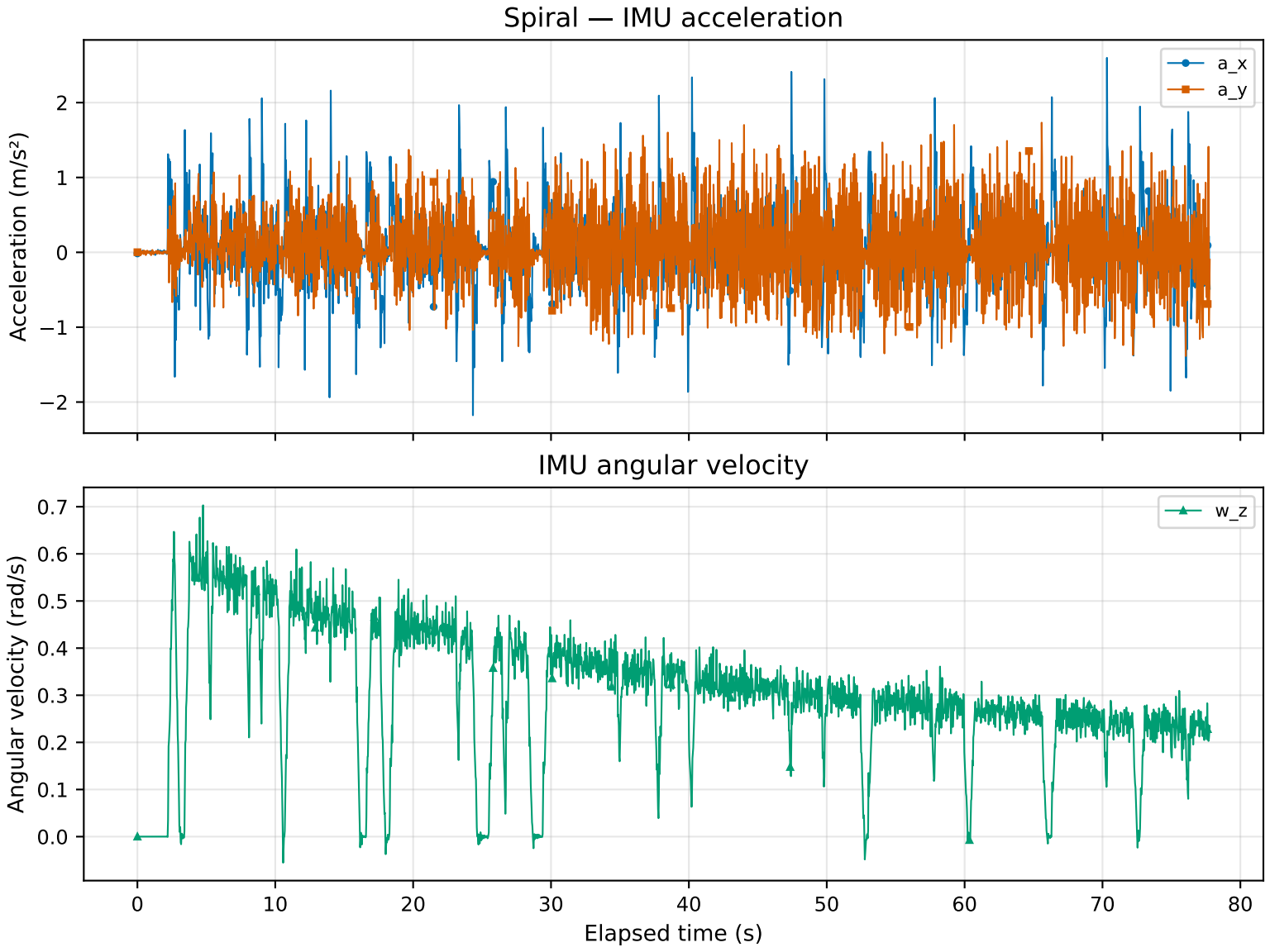
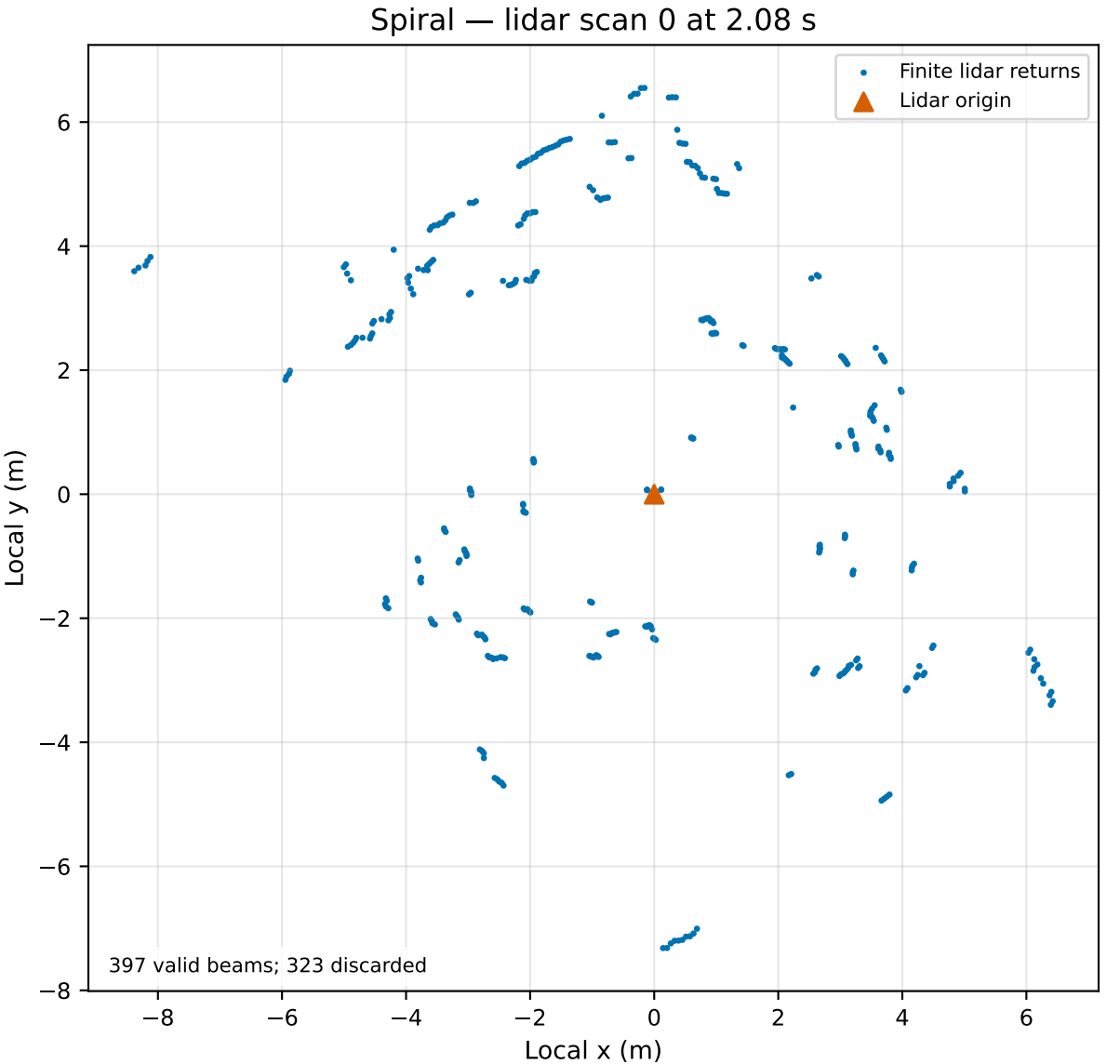

Spiral — odometry trajectory
Odometry path
Start
End
4.5
4.0
3.5
3.0
2.5
2.0
y (m)
−2.0
−1.5
−1.0
−0.5
0.0
x (m)


Spiral — position versus time
x
y
Position (m)
4
3
2
1
0
−1
−2
Heading versus time
Yaw (unwrapped)
Yaw (rad)
20
15
10
5
0
0
10
20
30
40
50
60
70
80
Elapsed time (s)


Spiral — IMU acceleration
a_x
a_y
Acceleration (m/s²)
2
1
0
−1
−2
IMU angular velocity
w_z
Angular velocity (rad/s)
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
10
20
30
40
50
60
70
80
Elapsed time (s)


Spiral — lidar scan 0 at 2.08 s
Finite lidar returns
Lidar origin
397 valid beams; 323 discarded
Local x (m)
Local y (m)
6
4
2
0
−2
−4
−6
−8
−8
−6
−4
−2
0
2
4
6

Lab 1 — Spiral sensor data

